

Noah Stokes

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EDUCATION

University of Waterloo
Bachelor of Computer Science

Sep. 2024 – May 2029
Waterloo, ON

EXPERIENCE

Software Engineering Intern
Government of Canada

May 2026 – Aug 2026
Ottawa, ON

- Unified search and completions into a single request flow for the agency-wide LLM chatbot, by re-architecting the backend from stateless chat completions to a session-based agentic pattern on Azure AI Foundry
- Let each query scope retrieval to a chosen subset of sources, by shipping a multi-source filtering layer over 5 connected data sources
- Returned partial transcripts in under 300ms with automatic English/French detection, by streaming real-time speech-to-text to the web client with the Azure Speech JavaScript SDK
- Caught regressions introduced during migrations by writing pytest integration tests across speech-to-text and retrieval filtering

Junior Software Developer
TUQ Inc.

Sep. 2025 – Dec. 2025
Waterloo, ON

- Cut completion time by ~65% for users filling out long medical forms, by shipping a full-stack conversational form assistant in Ruby on Rails and PostgreSQL that gathers information over multi-turn dialogue and auto-populates submissions
- Let users ask questions about pharmacy data, book appointments, and update records in plain language, by designing a PostgreSQL/pgvector retrieval layer and dispatching parsed requests to backing actions
- Reached 90% line coverage on core extraction and form-filling workflows by writing an RSpec suite and GitHub Actions CI pipeline with VCR-based HTTP mocking

PROJECTS

RL DevOps Autoscaler | *Python, PyTorch, Gymnasium, Stable-Baselines3, NumPy*

Aug. 2026

- Reduced resource-and-latency penalties by 24% versus traffic and CPU-based autoscaling baselines, by training a PPO agent to scale replicas under delayed provisioning
- Simulated 200–600 RPS workloads and 1–20 replicas with 18 automated tests, by modeling traffic, replica availability, queue buildup, p95 latency, and provisioning delays in Gymnasium

System Design Extension | *TypeScript, React, Python, FastAPI, LangChain*

Jul. 2026 – Present

- Answered architecture and system-design questions in VS Code, by building an agentic assistant extension that invokes a LangChain workflow to research the codebase, evaluate implementation options, and review designs
- Made long-running AI workflows interactive and persistent, by streaming progress, option cards, and Mermaid diagrams through a React webview while persisting per-workspace conversation state

Computer Use Agent | *Python, FastAPI, WebSockets, PyAutoGUI*

Jan. 2026

- Automated natural-language laptop tasks on connected machines, by converting requests into action plans executed through PyAutoGUI
- Adapted automations to UI changes and failed actions, by using screenshot feedback to verify state, retry actions, and dynamically replan

TECHNICAL SKILLS

Languages: Python, TypeScript/Javascript, Ruby, Java, SQL, C/C++

Frameworks & Libraries: React, Node.js/Express, FastAPI, Ruby on Rails, Flask, Vite

AI/ML: PyTorch, NumPy, Stable-Baselines3, Gymnasium, LangChain, RAG, vector search

Databases & Infrastructure: PostgreSQL, pgvector, ActiveRecord, Docker, Azure, GitHub Actions, CI/CD, Linux

Testing & Tools: Git, pytest, RSpec, VCR, REST APIs, WebSockets, unit & integration testing